

We would like to begin by thanking both referees and the editor for their thoughtful consideration and insightful feedback. We pursued all recommended avenues and address each point below. In total, with the direction you provided, we feel we have a robust paper, based on our original metric.

Following your suggestion, we made a significant revision by replacing the 2-Stage Least-Squares Model with a Stock-Flow model. We feel the paper benefits substantially from this enhanced reframing.

Reviewer 1

World Bank Data:

Your comment:

On page 2, still in the Introduction, the authors state, “For most of its history, the U.S. was less densely populated than its high-income peers, but in 1992 this changed. In that year, the U.S. grew to 28 people per square kilometer, outpacing the High Income Nations whose average was 23 people per square kilometer.” As you can see in the graph below, there was an extreme discontinuity in the High Income Nation data between 1991-1992.

In fact, in 1991, the density per square kilometer in High Income Nations was 34. I don’t know what happened to the data, but I don’t recall a global event of such significance that it would have decimated populations and/or destroyed enough land area in High Income Nations to result in the graph we see here. This gives me pause about using the World Bank data as your only source. I would encourage you to look for other data wherever you were previously relying solely on World Bank data. For what it is worth, the U.S. data from the World Bank does not exhibit this same pattern.

Our response:

We pulled fresh data from WorldBank which did not exhibit the same discontinuity. We revised our statement to,

“Between 2000 and 2022, population density in high-income countries increased by approximately 12 percent, while density in the United States increased by roughly 19 percent.”

Estimates of Housing Shortage:

Your Comment:

Your sentence in the first paragraph of the Introduction about the size of the housing unit shortfall in the US (4.4 million housing units). There is a good bit of debate about that number, so you might want to provide a range of estimates from a couple of reputable sources.

Our Response:

We agree and included a chart in the appendix, outlining recent estimates of housing shortage, noting their variance across time and magnitude.

Early Definition:**Your comment:**

Also on page 2, you define your Rental Density Index as “the quotient of population over rental units.” I almost did not read any further. If that were actually the definition, that would be fraught with too many issues to warrant further study. Fortunately, I pushed on, but it wasn’t really until page 7 that I got a better, cleaner definition. You need to make sure the reader gets to that information sooner, and do not muddy the proverbial waters with a loosely-worded definition that offers an incomplete/erroneous picture of what you are doing.

Our response:

We are grateful that you continued reading despite the poorly worded definition. We now introduce the proper definition in the abstract and reiterate it in the introduction before formally explaining the mathematics in Data and Descriptive Statistics.

Map Graphics:**Your Comment:**

I love a good map, but the maps provided on page 9 are not really visible (and therefore not useful in making your case).

Our Response: we agree, and we removed the map, as it was not helpful.

Large Home Builders' Perspective:**Your Comment:**

Part of the setup is an argument that “...when all available and affordable units are occupied, latent demand becomes invisible to market participants and researchers alike.” I disagree with that supposition. Multifamily developers routinely see through that cloak of invisibility. It would add a layer of richness to this study to talk to one or more large multifamily developers about how they analyze these same questions you are trying to address. They may have some insight that would really ratchet up the contribution of this piece. Alternatively, they may be impressed by the thoughtful and applied research you are doing that could directly impact their decision-making and bottom line. Either way, that is a win for the authors, in my view.

Our Response:

This comment was well taken, and we reached out to the homebuilding unit at Brookfield. Based on a discussion with their team, we bolstered the background section with reference to their methods and outlook.

Consideration of MSA-level Inflation

Your Comment:

Is there a way to come up with MSA-level inflation? I know you mention this in Section 3, but it would really help the analysis. Is there any way to back this out of any data the FHFA makes available?

Our Response:

We carefully considered the feasibility of constructing MSA-level inflation adjustments. We explored both FHFA metropolitan house price indexes and HUD Fair Market Rent series as potential candidates. FHFA indexes measure transaction prices in the owner-occupied housing market and are dominated by single-family sales, making them conceptually mismatched to rental rent growth in multifamily markets. HUD Fair Market Rents, while rental-based, are policy benchmarks designed for voucher administration; they incorporate smoothing, lagged information, and national CPI inputs and are not intended to measure year-to-year market rent inflation. Because neither source provides a consistent, market-based measure of local rental inflation over our sample period, we instead remove common inflationary and housing-cycle effects using year fixed effects and by normalizing rent growth relative to the national median each year. We clarify this choice in the revised manuscript.

Formatting Revisions:

Your Comment:

Also, that same paragraph highlights an issue with your references. I am sure it was just something that the program did when putting the references together, but it made some of the references hard to follow. For example, you frequently refer to “of Saint Louis” as an author and you cite “Bank, TW,” which I think is meant to refer to The World Bank. Make an editing pass and fix those issues with the references.

Our Response:

We revised these references

2SLS Reconsideration:

Your Comments:

Section 4.1, Page 12: “Individuals typically begin to form independent rental households in their early 20s. Therefore, the size of the birth cohort from 20 years ago directly influences the intensity of household formation pressure today.” Historically, this is absolutely true. Over the time period of your analysis, this is less- and-less the case every year. For this reason, I am not sold on the use of birth cohort size as a valid instrument. I also think that the current share of individuals born in the MSA 20 years prior is absolutely tied to concurrent housing affordability issues. If rents

recently spiked, that is most assuredly going to lead to an outmigration of 20-somethings. Are you controlling for rate of higher education attendance across MSAs? That will also impact your measured variable here.

What happens if the propensity to form independent households changes over time? It's been documented at length in the ownership literature that young people who lived through the Great Financial Crisis delayed housing purchases. What if there is a similar behavioral mechanism at work with rental household formation? Has anyone looked at that idea? I don't think it is a far-fetched concern that lingering effects from the GFC could be depressing the rate of new rental household formation. Perhaps look at a subset of cities hardest hit by the subprime mortgage fallout.

Two of your robustness checks are somewhat questionable, in my opinion. The placebo test that you report is essentially a tautology. In other words, it is of course the case that rents are going to respond to lagged RDI rather than contemporaneous RDI. Residential rents are set once per year, typically. Therefore, it will take a non-trivial amount of time for rents to catch up to market conditions. In a similar vein, I am not sure what the purpose of running a version of the instrument lagged by one year.

Our Response:

The reviewer brings up a number of strong doubts about the 2SLS model which caused us to reconsider this specification. We pivoted to a stock-flow model, rewriting the paper in accordance with that framework.

Diction:

Your comment:

Be very careful in your use of “outperform” and “underperform” when discussing rent increases. It is not entirely clear to me what those terms mean in this context.

Our response:

We revised, clarifying the definitions at first use.

Wording:

Your comment:

Page 19, Section 5.4, second paragraph – “Noteworthy is the 50th percentile on the x-axis, above which rent tends to be negative (5 of 8 years), and above which rent tends to be positive (7 of 9 years).” Maybe I am just misreading that sentence, but I am not following it at all.

Our response:

We revised and clarified.

Data Question on Vacant Dwellings:**Your Comment:**

Page 7 in the Data and Descriptive Statistics section: where are you getting data on vacant units and what is a seasonal dwelling in this context? Also, where does an AirBNB fall?

Our Response:

The IPUMS data include both occupied and vacant housing units, with vacancy status identified by the VACANCY variable. Our measures of renter household density and bedroom density are constructed using only renter-occupied households, and therefore exclude vacant and seasonal units by construction. An Airbnb unit would only be included if its owner reported it as their primary dwelling. We clarified this in the manuscript

Reviewer 2

Causes of RDI Movement:**Your Comment:**

Still in the Introduction on page 4, the authors state, “The reasons for an increasing RDI are, in most cases, population growth in excess of supply growth. However, it could be possible that housing units declined as populations remained static.” You need to identify any cases where that happened in your data, pull them out, and re-run your tests. Whether you fully report the results or just reference them in a footnote, you need to make sure it is as rare as you think and that it does not impact your results.

Our response:

Our statement was indeed reductive, and we should have fleshed it out. We revised with the following:

Although changes in RDI can arise mechanically through changes in either renter population or renter households, declines in rental unit counts are rare in our sample, occurring in fewer than 50 of 2,000 CBSA-year observations. And we explicitly control for supply in our model.

Explanation of RDI and its delta:

Your Comment:

Related to this discussion, the general idea you offer here is that RDI increases point to a greater latent rental demand but this behavior results in lower rents. Into a land of lower rents walks a multifamily developer to build an apartment complex. That makes little sense, and I would argue that it flies in the face of the Wheaton-DiPasquale Four-Quadrant model. I think some additional attention needs to be paid to the story/explanation here.

Our response:

We revised our framing based on this comment, shifting to a Wheaton-DiPasquale framework.

Writing Quality:**Your comments:**

I regret to say that the paper is very poorly written and edited, with mis-spellings, stray words, stray punctuation, missing words, false statements and meaningless verbiage scattered throughout.

Here are five instances in the first page or so:

1. Page 1: "Nearly half of U.S. renter households now spend over 30% of their income on housing". This is meaningless without context. Is that a lot?

2. " from 2000 to 2024, the consumer price index (CPI) for shelter exceeded the CPI for all other items by 30%". Perhaps the rate of increase is what is meant, but as written this does not make sense.

3. The citation "of St. Louis" as a last name shows a lack of care, as do later citations of "Mae" for Fannie Mae and other instances.

4. Page 2: "This juxtaposition of chronic national undersupply with localized oversupply underscores a deeper issue: the lack of a reliable metric to measure consumer housing demand". Does it? I am afraid the connection escapes me.

5. "absorption cannot exceed the rate of new deliveries" This is utterly false and Mueller and Laposa do not say this.

Our response:

We performed a thorough rewrite, reviewing, clarifying and specifying these comments and others.

Rent Growth and RDI in a stock-flow model:

Your Comments:

The most interesting and perhaps valuable part of the paper is the discussion of the relationship between rent growth and the level and change in the RDI. That discussion would benefit from a rigorous stock-flow discrete time model of the housing (rental) market which would capture adjustments to equilibrium in the manner of a error-correction model.

*The IV model discussion is very frustrating. If I understand correctly the IV is the percentage of the MSA's *current* population that was born in the same MSA 20 years ago. (THis is how they write it up, anyway). It would be nice if the authors told us how this was constructed. And do they mean *exactly* 20 years ago? That's oddly specific, and also before the usual age of household formation. But even then it's not a great instrument. The claim is that births 20 years ago have no relation to current shocks to the housing market, and that is true, I suppose. But that's not what the IV measures-- since it's the *percentage of the current population* it is clearly impacted by the number of people born 20 years ago who have left the MSA, as well as the number of people who entered the city, it is clearly not immune to current shocks.*

And the 2SLS model isn't right either if only because it treats current population and current housing stocks as exogenous, which definitely does not seem right.

Our Response:

Your questions and suggestions led us to reexamine our approach. Following your suggestion, we completely revised our framework and rewrote the draft, using a stock-flow model.

Cross-Sectional RDI

Your Comments:

On the RDI itself, we have this:

"The RDI itself is useful for identifying how tight a market is at a given point in time, and its absolute level is shaped by long-run demographic and structural trends in rentership and household formation. It is rare for a market to shift more than one half of one person-per-rental-dwelling, even over 20 years."

*This suggests that the cross-section variation of RDI is going to be more interesting and important than within-city changes in the RDI. And it *is* interesting, although not for reasons that the authors consider*

*The authors emphasize the *change* in RDI and I found that puzzling for two reasons. The first is that the above quote would seem to suggest that changes don't have a lot of variation, and the second is the statement that the number of households rises monotonically. They might mean the total number of households in this context but that whole paragraph is a bit obscure.*

Our Response:

In the revised draft, we clarify that while the absolute level of the RDI is influenced by slow-moving demographic and structural factors, economically meaningful valuation arises both across cities and within cities over time; we have revised the text to avoid implying that cross-sectional variation is inherently more informative. We now more clearly distinguish between the structural determinants of the index's level and the cyclical or policy-relevant movement that occur within markets.

Omitted considerations:

Your comment:

"Within the 200 MSAs in the study, the RDI ranges from 1.68 (Bloomington, IN; 2023) to 3.74 (Provo, UT; 2009)"

which suggests not so much housing market pressure, but family size choices. More needs to be said about that contrast. The author ignores the age distribution of the renter population, but this is clearly important for the RDI.

As a side note, but an important one, I think, a better metric would be renters per bedroom, or renters per square feet. This would capture true crowding (although the age distribution would still need accounting) and would also help control for the shift to single family rentals, which is ongoing in this period and which is not discussed at all by the authors.

Our Response:

We were very interested in this comment and revised the entire model to focus instead on renters per bedroom.